Bank Marketing Subscription Prediction

(Final)

Team 12

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Outline

1. Introduction
2. Problem Setting
3. Proposed Methods
4. Experiments Setting
5. Experiments Results
6. Conclusion and Limitation
7. Work distribution chart

Outline

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   - 1.4. Challenges
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Background

- Marketing campaigns are crucial in the banking industry.
- **Term deposits** are an important product that provides stable funding for banks.
- Traditional marketing strategies are costly and often achieve low conversion rates.
- **Data-driven approaches** can improve targeting, reduce expenses, and enhance profitability.



Motivation

1. Traditional marketing is inefficient → few subscriptions
2. Data-driven strategy improves ROI
3. Compare multiple models to understand performance trade-offs between accuracy and interpretability.
4. Apply interpretable insights to support marketing strategy refinement and customer targeting.

Research Aims

1. Build a **predictive model** for term deposit subscription (Yes/No)
2. Explore **feature** importance for interpretability and marketing insights.
3. Compare algorithms
4. Ensure robust generalization (stratified k-fold CV)
5. Benchmark vs Kaggle baselines & related works

Challenges

1. Class Imbalance Problem
    - There exists a significant imbalance between positive and negative classes.
    - To mitigate bias, we apply resampling techniques such as SMOTE on the training set only and evaluate the model using AUROC and F1-score to ensure balanced performance.
2. Overfitting Risk and Model Generalization
    - Complex models are prone to overfitting. We address this by applying regularization and early stopping during training.
    - Cross-validation is further used not as a prevention, but as a method to assess model robustness and generalization consistency.

Contributions

1. Addressed the two major challenges in this project: class imbalance and overfitting.
2. Used SHAP to identify **call duration** as the most influential feature across all models.
3. Conducted a realistic “no-duration” experiment: removing duration caused a 27% drop in F1-score, confirming dependence on in-call information.
4. Found that **balance** becomes the strongest predictor when duration is unavailable, providing actionable pre-call targeting insights.
5. Identified **specific months** (March, September, October, December) with higher subscription success rates, offering practical guidance for campaign scheduling.
6. Explained LightGBM’s superiority through its hybrid use of interaction features and raw features.

Outline

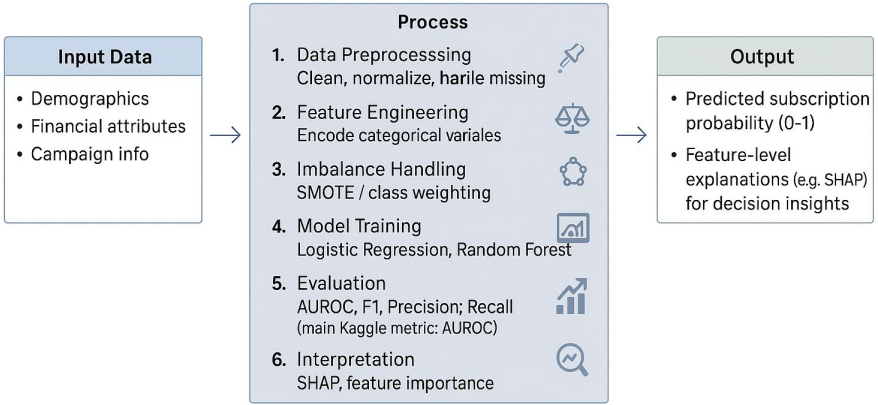
1. Introduction
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   - 2.2. Evaluation Metrics
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Problem Description

1. Input
   a. Demographics, financial attributes, campaign info

2. Process:
   a. **Data Preprocessing**: Clean, normalize, handle missing values
   b. **Feature Engineering**: Encode categorical variables
   c. **Imbalance Handling**: SMOTE / class weighting
   d. **Model Training**: Logistic Regression, Random Forest, CatBoost
   e. **Evaluation**: AUROC, F1, Precision, Recall
   f. **Interpretation**: SHAP, feature importance

3. Output:
   a. Predicted subscription probability (0–1)
   b. Feature-level explanations for decision insights



Problem Description

Research Problem:

- This study aims to develop and evaluate interpretable machine learning models for predicting bank clients' subscription behavior.
- By addressing class imbalance and enhancing model explainability, the project seeks to improve targeting accuracy, reduce marketing costs, and provide actionable insights into key behavioral and demographic factors influencing client decisions.

Evaluation Metrics

Metric	Description	Purpose
Accuracy	Overall percentage of correct predictions.	General baseline indicator.
Precision	Proportion of predicted positives that are true positives.	Measures false-positive control.
Recall (Sensitivity)	Proportion of true positives that are correctly detected.	Important for minority class detection.
F1-score	Harmonic mean of Precision and Recall.	Balances precision–recall trade-off for imbalanced data.
AUROC	Area under the ROC curve.	Primary metric for Kaggle; robust to class imbalance.

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   - 3.2. Preprocessing Steps
   - 3.3. Medeling Strategy
   - 3.4. Implementation Plan
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Dataset Overview

1. Dataset:
    - Kaggle Playground S5E8 (Bank Marketing Subscription)
    - https://www.kaggle.com/competitions/playground-series-s5e8/overview
2. Source:
    - Synthetic version of UCI Bank Marketing (Moro et al., 2014)
3. Size: 750k train, 250k test
4. Features:
    - Categorical: job, marital, education, housing, loan, contact type, etc.
    - Numerical: age, balance, duration, previous contacts, etc.
5. Target: subscribed (binary Yes/No)

Key Observations

- No missing values, all columns clean and consistent.
- Highly imbalanced target: positive (≈11%) vs. negative (≈89%).
- Strongest signal: duration positively correlated with subscription.
- Most categorical variables weakly correlated, implying non-linear patterns.

Preprocessing Steps

- Convert categorical variables to category
  - For efficient memory and encoding
  - Solve challenge: Model Generalization

- Normalize / log-transform skewed numeric variables
  - balance_log, duration_sin/cos, pdays_sin
  - Stabilize distribution
  - Solve challenge: Overfitting Risk and Model Generalization

Preprocessing Steps

- Create interaction features (numeric bigrams)
  - Capture non-linear effects between variables.
  - Solve challenge: Weakly Correlated

- Apply Target Encoding (TE) and Count Encoding (CE)
  - Uing both competition and original UCI dataset
  - Enriched categorical representations.
  - Solve challenge: Class Imbalance

Preprocessing Steps

- Ensure no data leakage
  - Transformations using y (e.g., TE, SMOTE) are fitted on training folds only
  - External statistics (e.g., CE from orig) are computed offline without target labels.
  - Solve challenge: Overfitting & Generalization

Modeling Strategy

- Explore both tree-based and deep tabular architectures
  - Ensure diversity and robustness.
- Use Stratified 5-fold Cross-Validation for consistent evaluation across all models.
- Optimize hyperparameters with early stopping.
  - learning rate, embedding dim, dropout, epochs

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Data Description and Characteristics (review)

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#	△	Team	Members	Score	Entries	Last	Solution
1	—	Optimistix		0.97801	132	2mo	
2	▲ 3	Mahog		0.97794	60	2mo	
3	▲ 1	Succeeded. Resource recovered		0.97790	145	2mo	
4	▼ 2	Masaya Kawamata		0.97790	82	2mo	
5	▼ 2	Ravi Ramakrishnan		0.97784	28	2mo	
6	▲ 9	Kyr1ll		0.97775	31	2mo	
7	▼ 1	wind1234it		0.97769	106	2mo	
8	▲ 3	DanteTheAbstract		0.97768	31	2mo	
9	▲ 1	Yeshua T		0.97768	25	2mo	
10	▼ 1	Thiago Lima Santos		0.97767	45	2mo	

Dataset Checklist

1. Domain:
   - ✓ **Finance & Economics**
2. Source:
   - ✓ **Public** Kaggle competition (leaderboard & baselines available)
3. Size:
   - ✓ **1M** samples (≫ 5k minimum)
4. Features:
   - ✓ Clear semantics (age, job, contact, etc.) → interpretable
5. Task:
   - ✓ **Binary classification** (Yes/No subscription)

Basic Exploratory Data Analysis (EDA)

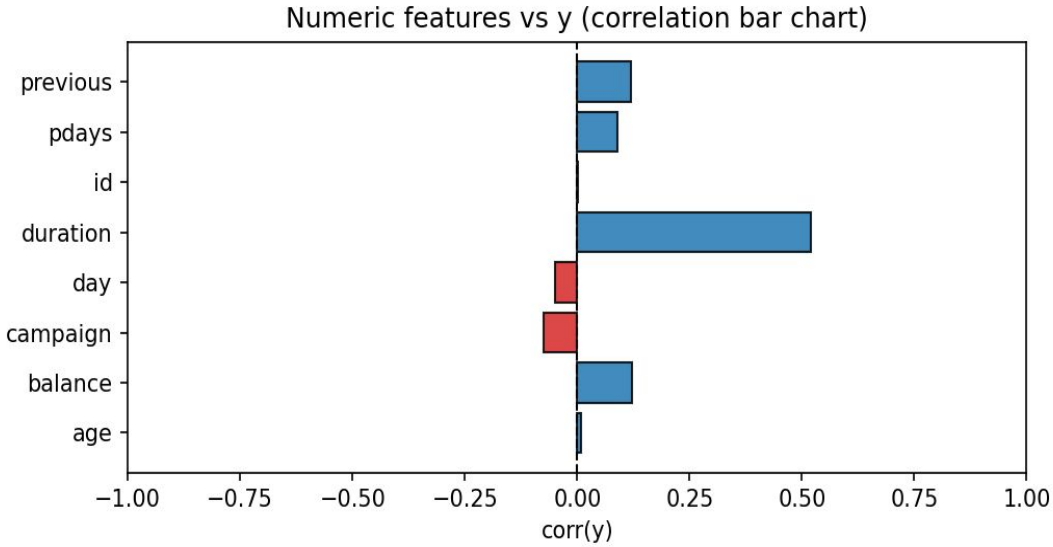
Variable	Description	Type / Values
age	Client’s age	Numeric
balance	Average yearly balance (EUR)	Numeric
day	Last contact day of month	Numeric (1–31)
duration	Last contact duration (seconds)	Numeric
campaign	Number of contacts in this campaign	Numeric
pdays	Days since last contact (-1 = never contacted)	Numeric
previous	Number of contacts before this campaign	Numeric
y	Subscribed to term deposit? (target)	Binary: yes, no

Basic Exploratory Data Analysis (EDA)

Variable	Description	Type / Values
job	Type of job	Categorical: admin., blue-collar, entrepreneur, etc.
marital	Marital status	Categorical: married, single, divorced
education	Education level	Categorical: primary, secondary, tertiary, unknown
default	Has credit in default?	Categorical: yes, no
housing	Has a housing loan?	Categorical: yes, no
loan	Has a personal loan?	Categorical: yes, no
contact	Contact communication type	Categorical: unknown, telephone, cellular
month	Last contact month	Categorical: jan, feb, …, dec
poutcome	Outcome of previous campaign	Categorical: unknown, other, failure, success

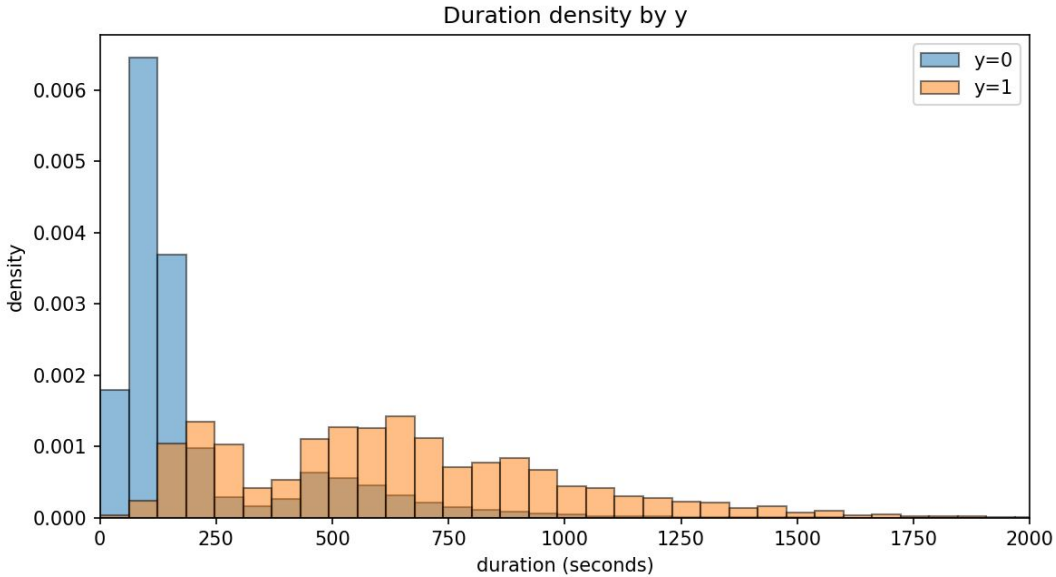
Basic Exploratory Data Analysis (EDA)

1. missing value rate: 0%
2. numeric vs y correlation analysis
   - duration is strongest signal



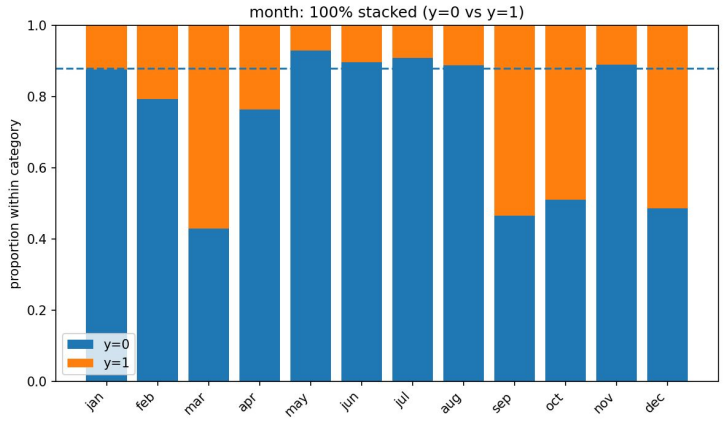
Basic Exploratory Data Analysis (EDA)

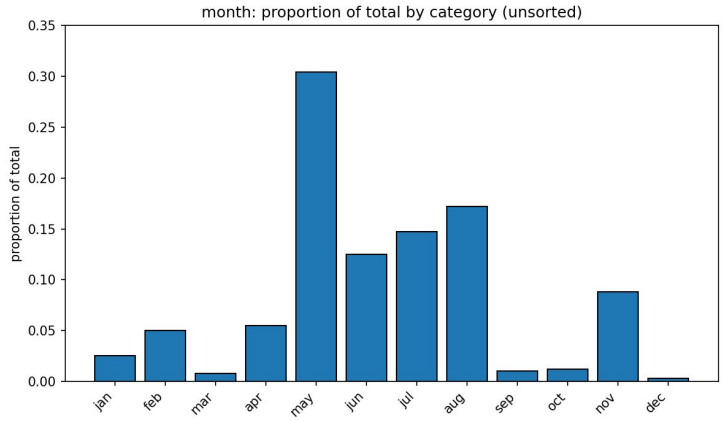
- numeric vs y correlation analysis (duration)
  - Interested clients talk longer



Basic Exploratory Data Analysis (EDA)

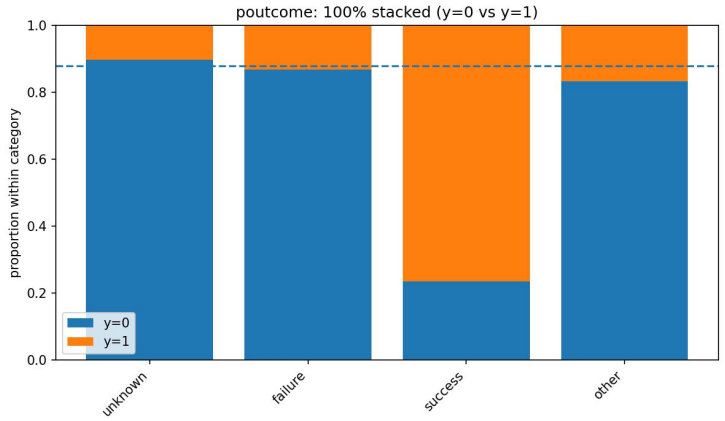
- Analyze non-numeric data (month)
  - In Mar, Sep, Oct, Dec, the proportion of true is high
  - However, their overall share of data is relatively low

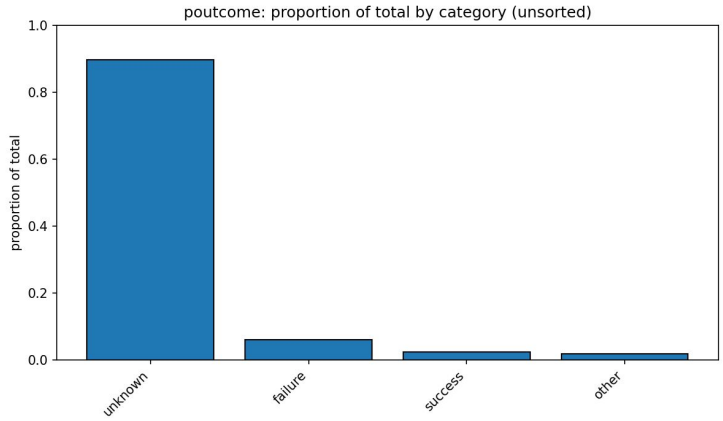




Basic Exploratory Data Analysis (EDA)

- Analyze non-numeric data (poutcome)
  - If the client had a previous successful outcome,
  - The probability of success is much higher
  - However, this category has very few samples





Basic Exploratory Data Analysis (EDA)

- Summary:
  - No missing data
  - Most categories show no clear correlation with y
  - Correlated categories cover only a small portion of data

Basic Exploratory Data Analysis (EDA)

- Dataset clean and usable → feasible for modeling
- Low correlation → signals are non-linear patterns
- Plan:
    a. Create interaction & composite features
    b. Use ensemble models for hidden relationships
    c. Apply SHAP for interpretability

Synthetic vs. Original Dataset

- Purpose:
  - To expand the training dataset and improve model generalization.
- Difference:
  - Generated by a deep model, so feature distributions are similar but not identical.
  - May alter rare patterns or correlations among variables.
- Potential Distortion:
  - Synthetic process may introduce bias or loss of realism, especially in edge cases.

Baseline Methods

Baseline – AdaBoost

- Classic boosting method combining weak learners.
- Simple, interpretable, and commonly used in telemarketing prediction.
- Limited in capturing nonlinear patterns and performs poorly on imbalanced data.

SOTA – RNN

- Deep learning model with hidden-state memory for sequential feature modeling.
- Tabular features reshaped into pseudo-sequences to learn feature interactions.
- Achieved the best overall performance, especially in Recall and F1-score.

Evaluation Strategy

- Use Stratified 5-Fold CV
- Evaluate each fold independently and compute OOF (Out-of-Fold) scores
- Compare cross-model results (Adaboost, RNN, LGBM) using the same metrics.
- Evaluation metrics: AUROC, F1-score, Accuracy, Precision and Recall.
  - Solve Challenge: Class Imbalance

Environment

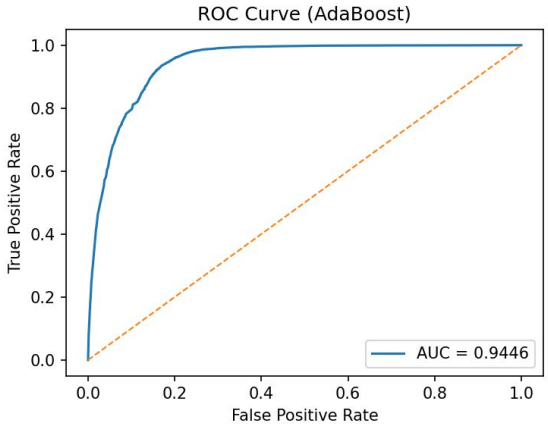
1. OS
    - Linux / Windows
2. Programming Language
    - Python
3. Tools
    - VSCode
4. Libraries
    - pandas, numpy, scikit-learn, tensorflow, lightgbm

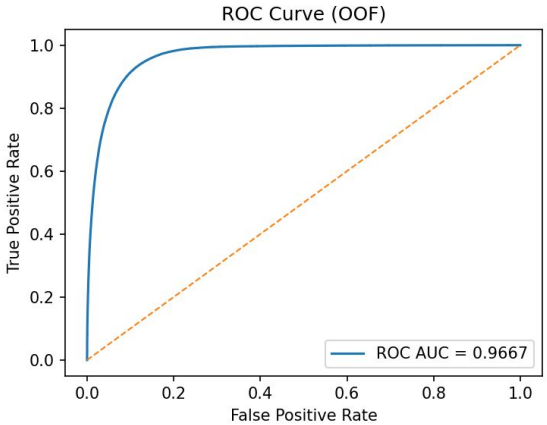
Outline

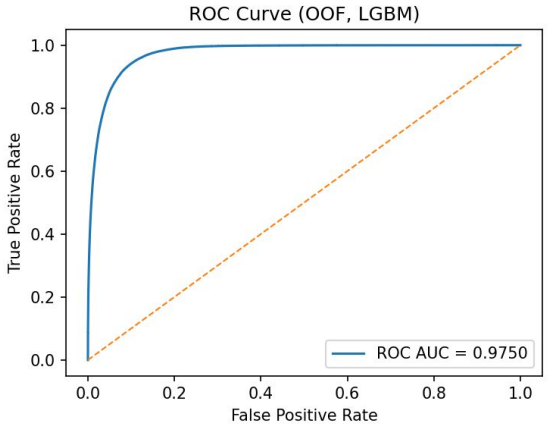
1. Introduction
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5. Experiments Results
   - 5.1. Comprehensive experimental results (including some texts/figures)
   - 5.2. Please describe your insights or thoughts based on the results.
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7. Work distribution chart

ROC Curve Comparison

- AdaBoost < RNN < LGBM

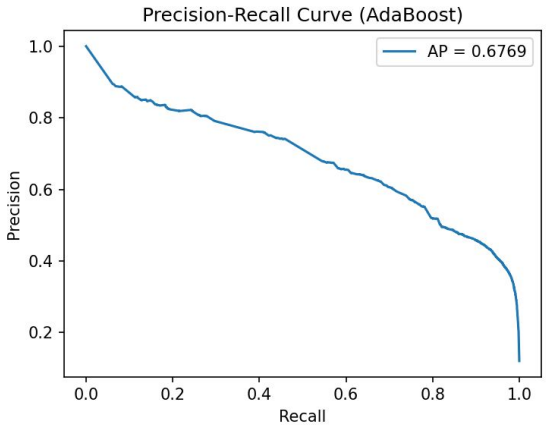


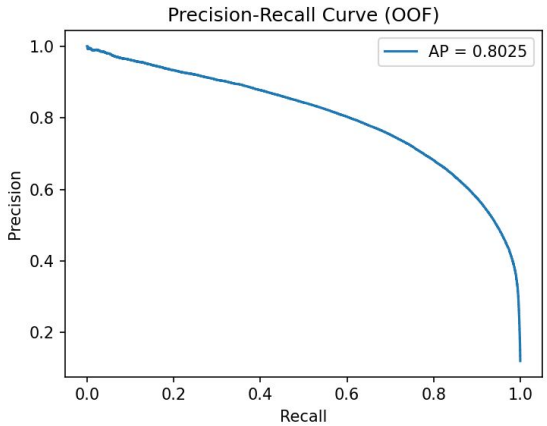


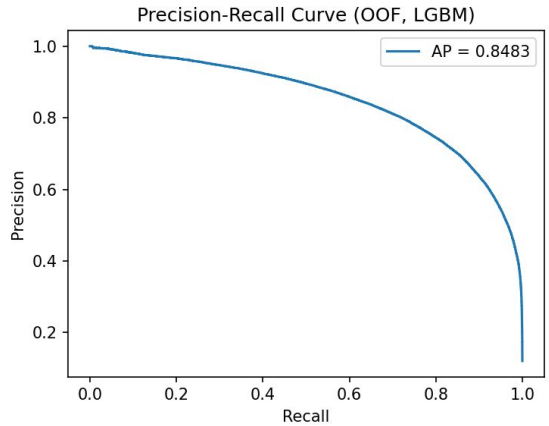


PR Curve Comparison

- AdaBoost < RNN < LGBM

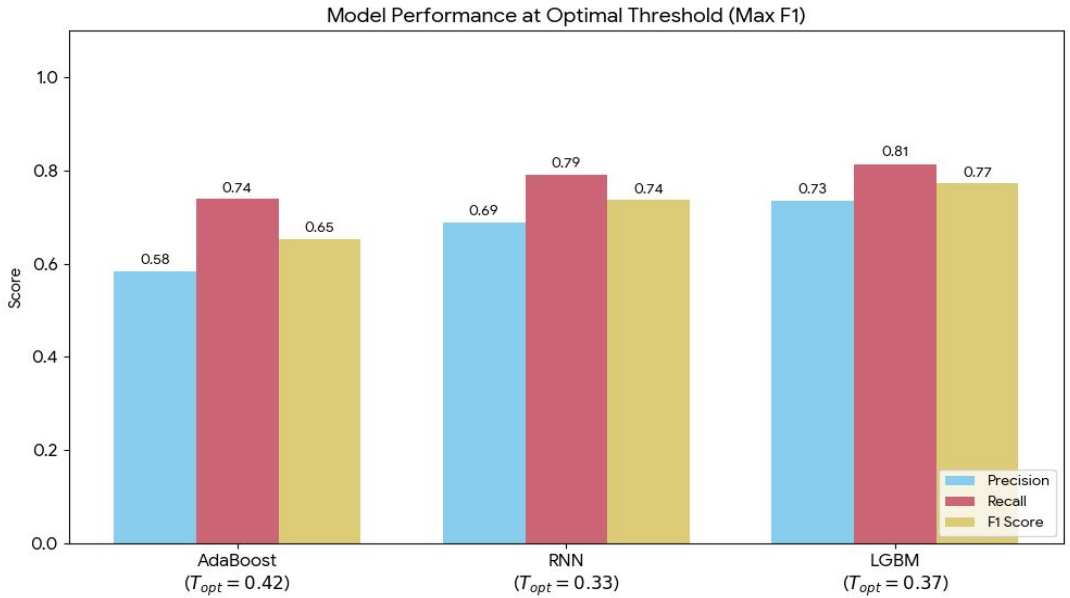






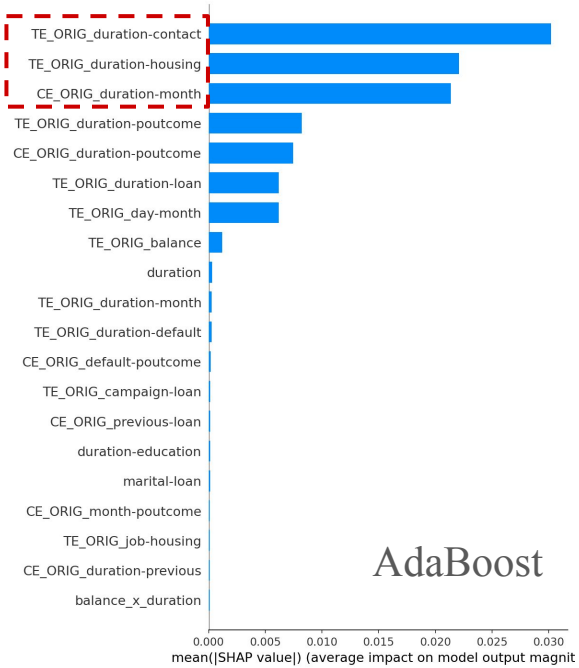
Optimal Threshold & F1 Analysis

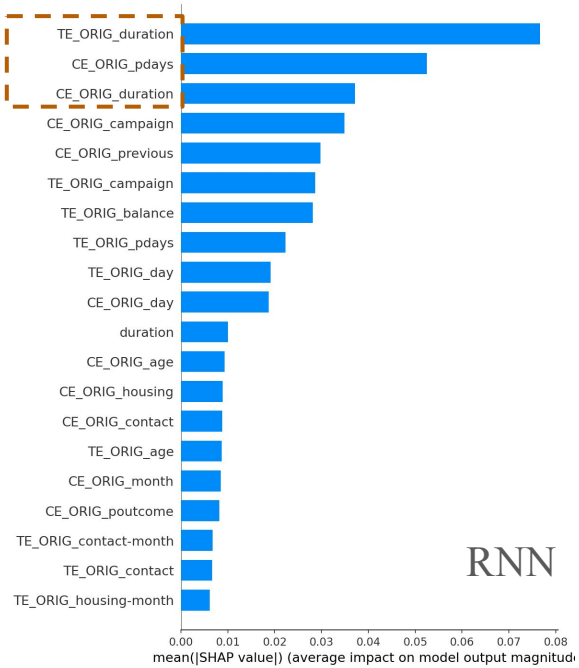
- AdaBoost < RNN < LGBM

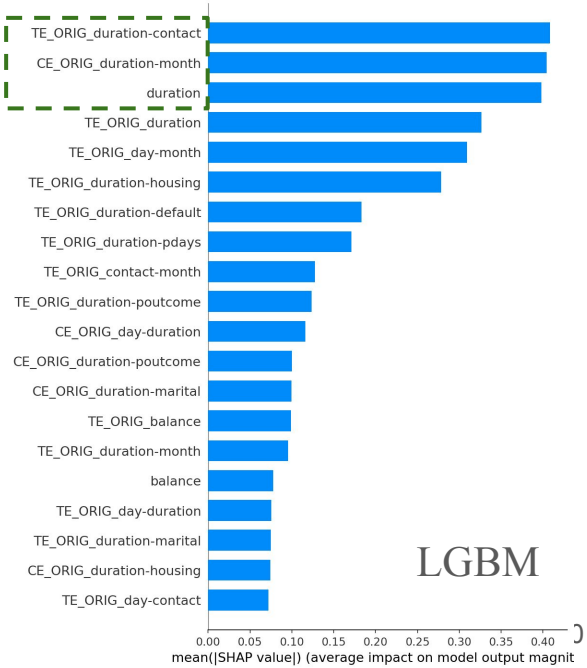


SHAP Feature Importance

- LGBM excels by balancing raw signals and interactions

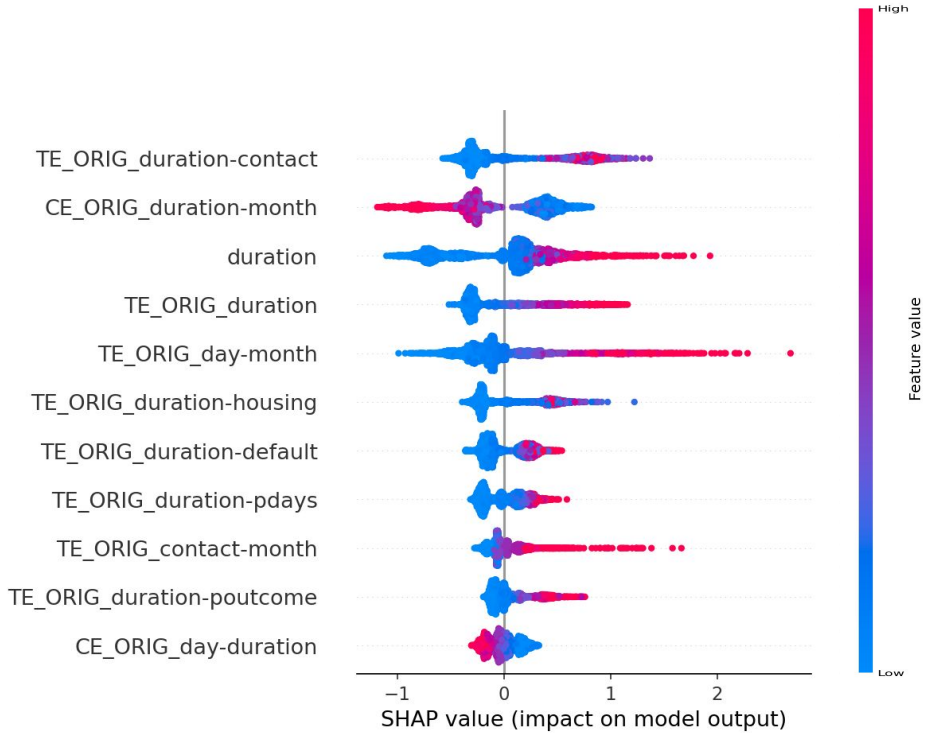






Decoding the Strategy

- Impact Direction
  - Higher values (Red), higher subscription probability
- Discrimination Power
  - Clear separation between potential subscribers and non-subscribers



Performance Summary Table

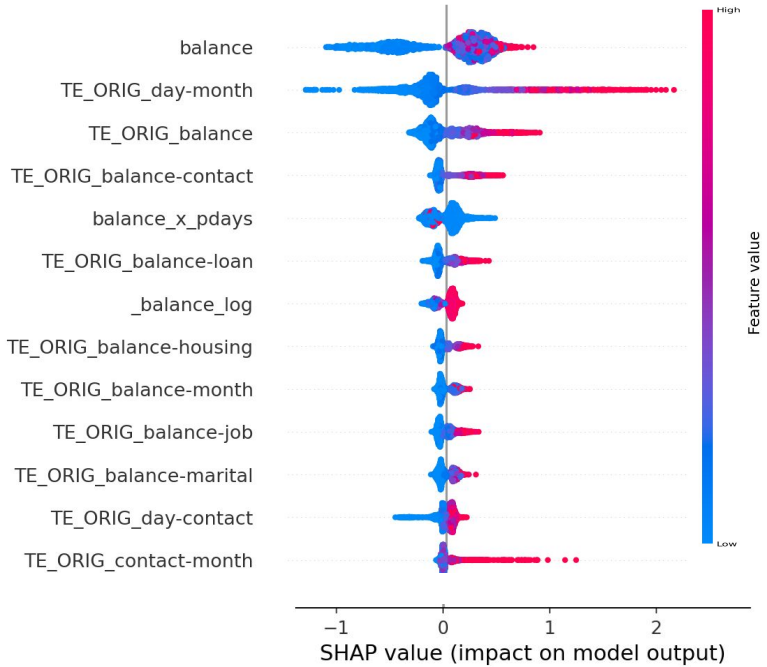
Model	F1-opt	Threshold	Precision	Recall	ROC-AUC	AP (PR-AUC)
AdaBoost	0.6517	0.421	0.5832	0.7384	0.9446	0.6769
RNN	0.7365	0.331	0.6888	0.7912	0.9667	0.8025
LightGBM	**0.7721**	0.368	**0.7345**	**0.8137**	**0.9750**	**0.8483**

Realistic Scenario: Predicting Without Call Duration

Metric	Original	**Without Duration**	Diff
F1 Score	0.7721	**0.5641**	↓ 27%
Threshold	0.368	**0.268**	↓ 0.1
ROC-AUC	0.9750	**0.7844**	↓ 0.19
AP (PR-AUC)	0.8483	**0.5404**	↓ 0.3

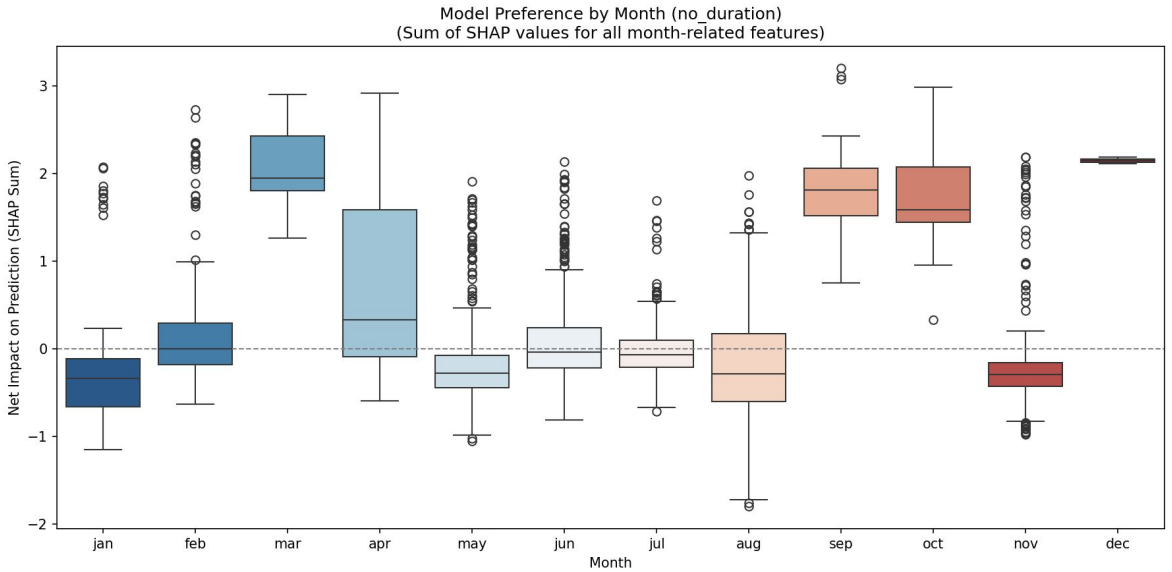
Strategic Pivot: What If We Don't Know Duration?

- Insight: Financial Power is the New King
  - Without call duration, Balance becomes the #1 predictor.
- Action: Prioritize "High Balance" Leads
  - Filter and rank calling lists by customer's financial depth before dialing.



Strategic Pivot: What If We Don't Know Duration?

- Action: Prioritize "High Balance" Leads
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Insight

- Low Impact:
  - Static Demographics: Job, Marital, Education
  - Generic Loan Flags: Housing, Loan
  - Insight: Don't waste marketing resources on filtering lists simply based on "occupation"
- High Potential:
  - Financial Flow: Payday, large fund transfers
  - Digital Intent: Fixed deposit page browsing and interest rate calculation records.
  - Detailed feedback: A detailed code indicating the reason for rejection

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Conclusion

- Built an accurate and interpretable prediction framework
- Effectively addressed imbalance, weak signals, and overfitting
- LGBM wins via hybrid feature strategy
- Success needs context, not just duration
- Target high balance in peak seasons

Limitation & Future

- Removing duration drops performance by 27%
- Static demographics lack real predictive power
- Future: Integrate dynamic financial flow data

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Work Distribution Chart

Job	Member
Paper survey	陳靖燁
EDA	白宇辰
Experiments	陳靖燁、白宇辰

Reference

1. https://www.kaggle.com/competitions/playground-series-s5e8/overview
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3. https://www.sciencedirect.com/science/article/abs/pii/S016792361400061X
4. https://archive.ics.uci.edu/dataset/222/bank+marketing

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